

Week 2 Assignment: Black-Scholes Pricing

TO: Portfolio Manager
FROM: Ramit Goyal (Options Analyst)
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SUBJECT: Black-Scholes Pricing

1 Underlying Asset Selection

We are analyzing equity options for **Reliance Industries Limited (RELIANCE)** using data captured directly from the official National Stock Exchange of India (NSE) website. Comparing their real market premiums to European pricing highlights the behavior of BS equation here.

Below is the live reference snapshot of the NSE option chain used :

- **Spot Price (S_0):** 1,321.90 (Sourced from above : Underlying Symbol: RELIANCE 1,321.90).
- **Strike Price (K):** 1,320.00 (The closest listed option strike row relative to the current spot price).
- **Time to Maturity (T):** The selected contract expiry date is June 30, 2026. Given the data extraction timestamp of May 29, 2026, there are precisely 32 calendar days remaining until expiry. Converting to years :

$$T = \frac{32}{365.25} = \mathbf{0.08761} \text{ years} \quad (1)$$

2 Volatility Estimation (σ)

We calculate it using two methodologies:

1. **Historical Volatility (σ_{hist}):** Calculated by taking the standard deviation of daily log transformed closing price returns over the previous 30 trading days :

$$\sigma_{\text{hist}} = \text{Standard Deviation}(\ln(P_t/P_{t-1})) \times \sqrt{252} = \mathbf{21.50\%} \quad (\mathbf{0.2150}) \quad (2)$$

2. **Market Implied Volatility (σ_{implied}):** Extracted directly from the **IV** data column within our targeted 1,320.00 strike price:

- Call side Implied Volatility: **20.05%** (**0.2005**)
- Put side Implied Volatility: **25.87%** (**0.2587**)

The Volatility Difference:

The standard Black-Scholes model assumes that volatility is exactly the same for every single option contract on the same stock. However, our real world market data shows that this isn't true. Instead, we see a clear pattern due to distinct volatility patterns.

3 Risk Free Rate (r)

For options clearing on the NSE, the standard benchmark proxy for a risk-free return rate is currently trading at 6.50% annualized. Because the analytic Black-Scholes framework operates assuming continuous compound mechanics, we mathematically transform the standard nominal rate via a natural logarithm:

$$r = \ln(1 + 0.065) = \ln(1.065) \approx \mathbf{0.06297} \implies \mathbf{6.30\%} \quad (3)$$

4 Black-Scholes Evaluation

We feed our filtered parameter list directly into the mathematical pricing script :

$$S = 1321.90, \quad K = 1320.00, \quad T = 0.08761, \quad r = 0.0630, \quad \sigma = 0.2005$$

4.1 Step 4.1: Computing Components

First, we solve for :

$$\sigma\sqrt{T} = 0.2005 \times \sqrt{0.08761} = 0.2005 \times 0.2960 = \mathbf{0.05935} \quad (4)$$

Next, we calculate the continuous log ratio of S and K :

$$\ln\left(\frac{S}{K}\right) = \ln\left(\frac{1321.90}{1320.00}\right) = \ln(1.001439) = \mathbf{0.00144} \quad (5)$$

Then, we solve for our continuous risk adjustment coefficient:

$$\left(r + \frac{\sigma^2}{2}\right) T = \left(0.0630 + \frac{0.2005^2}{2}\right) \times 0.08761 = (0.0630 + 0.02010) \times 0.08761 = \mathbf{0.00728} \quad (6)$$

4.2 Step 4.2: Calculating d_1 and d_2 Distributions

Using the solved components, we compute our normalized probability positions:

$$d_1 = \frac{\ln(S/K) + \left(r + \frac{\sigma^2}{2}\right) T}{\sigma\sqrt{T}} = \frac{0.00144 + 0.00728}{0.05935} = \frac{0.00872}{0.05935} = \mathbf{0.14692} \quad (7)$$

$$d_2 = d_1 - \sigma\sqrt{T} = 0.14692 - 0.05935 = \mathbf{0.08757} \quad (8)$$

4.3 Step 4.3: Mapping Normal Probability Density Functions $N(\cdot)$

We map our calculated values onto a cumulative standard normal distribution matrix. This step converts our coordinates into probabilities:

- $N(d_1) = N(0.14692) = \mathbf{0.55841}$ (Delta)
- $N(d_2) = N(0.08757) = \mathbf{0.53489}$ (The statistical risk neutral probability of expiration inside-the-money)
- $N(-d_1) = 1 - 0.55841 = \mathbf{0.44159}$
- $N(-d_2) = 1 - 0.53489 = \mathbf{0.46511}$

4.4 Step 4.4: Computing Final Asset Options Value

We calculate our continuous currency discount factor:

$$e^{-rT} = e^{-(0.0630 \times 0.08761)} = e^{-0.00552} = \mathbf{0.99449} \quad (9)$$

1. Theoretical European Call Value (C):

$$\begin{aligned} C &= S \cdot N(d_1) - K \cdot e^{-rT} \cdot N(d_2) \\ C &= (1321.90 \times 0.55841) - (1320.00 \times 0.99449 \times 0.53489) \\ C &= 738.161 - (1312.727 \times 0.53489) \\ C &= 738.161 - 702.164 = \mathbf{36.00} \end{aligned} \quad (10)$$

2. Theoretical European Put Value (P):

$$\begin{aligned} P &= K \cdot e^{-rT} \cdot N(-d_2) - S \cdot N(-d_1) \\ P &= (1320.00 \times 0.99449 \times 0.46511) - (1321.90 \times 0.44159) \\ P &= (1312.727 \times 0.46511) - 583.738 \\ P &= 610.533 - 583.738 = \mathbf{26.79} \end{aligned} \quad (11)$$

5 Market Calibration

We map our closed form Black-Scholes metrics against the live Last Traded Prices (**LTP**) shown on the NSE chain data table:

Option Contract (1,320 Strike)	Model Value	Live Market LTP	Absolute Spread Variance
Call Option Premium	36.00	133.50	+97.50 (+270.8%)
Put Option Premium	26.79	3.70	-23.09 (-86.2%)

Table 1: Analytical Model Output vs. Exchange-Traded Order Book Values

Deconstructing the Pricing Discrepancy Framework:

Evaluating the structural spread variables exposes a complete decoupling from textbook theory. Because our engine evaluated both contracts using a symmetric volatility parameter (20.05%), the system output fails to align with the real world order book prices:

- The Call Premium Expansion:** The market call option is trading at a premium of 133.50, heavily outpacing our theoretical calculation of 36.00. This structural anomaly indicates that aggressive buyers are pricing in a highly positive forward momentum event, such as an upcoming dividend declaration or an expected trend reversal. This buying momentum inflates the option price way past normal log normal distributions.
- The Put Compression Asset Gap:** Conversely, the real exchange put option is trading down at 3.70, way below our calculated baseline model price of 26.79. This extreme compression shows that despite the standalone calculated IV, the broader market expects an incredibly low probability of Reliance dropping below 1,320 before expiration, keeping the option price compressed.

6 The Greeks

The Greeks define partial derivatives across our model variables, showing how our option price shifts when market conditions change:

- **Delta** ($\Delta = 0.5584$): Directional sensitivity. For every 1.00 move upward in Reliance stock, our call option price will gain 0.56.
- **Gamma** ($\Gamma = 0.0050$): Speedometer for Delta. If the stock goes up by 1.00, our Delta will grow from 0.5584 to 0.5634.
- **Vega** ($\nu = 1.52$): Volatility sensitivity. If the market implied volatility jumps by 1% absolute, our call price will gain 1.52.
- **Theta** ($\Theta = -0.58$ **per day**): Time decay. Every 24 hours that pass strip 0.58 from our option value just because time is running out.

6.1 Risk Diagnostics Portfolio Example

If our underlying stock increases by 1% (+13.22) concurrent with an absolute 1% spike inside the asset implied volatility profile (+1%):

$$\Delta C \approx (\Delta \times \Delta S) + (\nu \times \Delta \sigma\%) \quad (12)$$

$$\Delta C \approx (0.5584 \times 13.22) + (1.52 \times 1) = 7.382 + 1.520 = +\mathbf{8.90} \quad (13)$$

Our adjusted forward theoretical call value estimate equals: $36.00 + 8.90 = \mathbf{44.90}$.

7 American Early Exercise

NSE single stock options allow for early exercise before the expiration date. For our **call option**, this early exercise feature adds zero premium value because exercising early strips away the remaining time value of the option and ties up capital unnecessarily.

For our **put option**, early exercise can have real value if the option goes deep in-the-money ($S \ll K$), because it lets the holder get cash immediately and start earning interest. However, since the Reliance spot price (1,321.90) is currently above our strike price (1,320.00), the option carries zero intrinsic value ($1320 - 1321.90 < 0$). This means the early exercise value right now is zero, making its impact on our pricing discrepancy very small.

8 Analytical Reflection

Applying the Black-Scholes model to real market data revealed a clear gap between clean textbook physics and messy market realities. In theory, equations assume smooth mathematical distributions and static variables. In practice, seeing how much real-world call and put prices deviate from our mathematical outputs proved that markets do not follow a perfect bell curve. This exercise showed me that option premiums are driven by real human behaviors, which cannot be captured by a formula alone. It turns option pricing from a simple math problem into a continuous challenge of modeling dynamic boundaries.

Appendix: Clean Python Verification Script

```
import numpy as np
from scipy.stats import norm

# Inputs from the real NSE screenshot
S = 1321.90
K = 1320.00
T = 0.08761
r = 0.0630
sigma = 0.2005

# Pricing Calculation
d1 = (np.log(S / K) + (r + 0.5 * sigma**2) * T) / (sigma * np.sqrt(T))
d2 = d1 - sigma * np.sqrt(T)

call_price = S * norm.cdf(d1) - K * np.exp(-r * T) * norm.cdf(d2)
put_price = K * np.exp(-r * T) * norm.cdf(-d2) - S * norm.cdf(-d1)

print(f"--- BLACK-SCHOLES PIPELINE ---")
print(f"Computed d1: {d1:.5f} | d2: {d2:.5f}")
print(f"Theoretical Call Price: Rs. {call_price:.2f}")
print(f"Theoretical Put Price: Rs. {put_price:.2f}")
print(f"Calculated Call Delta: {norm.cdf(d1):.4f}")
```

References

- [1] Zerodha Varsity. *Options Theory - The Black-Scholes-Merton Pricing Engine*. <https://zerodha.com/varsity/module/options-theory/>
- [2] Investopedia. *Protective Put Options Strategy Guide*. <https://www.investopedia.com/terms/p/protective-put.asp>